Name: ______________________________ Period: ________ Date: ________

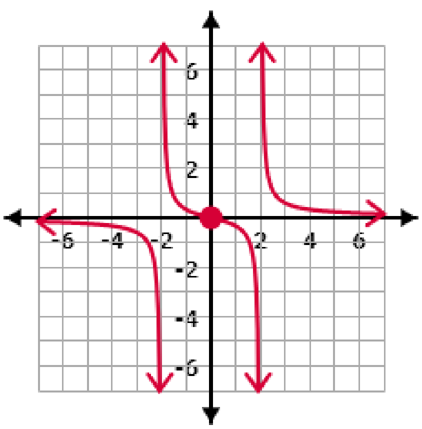
OnRamps Precalculus
Unit 5 Test Review

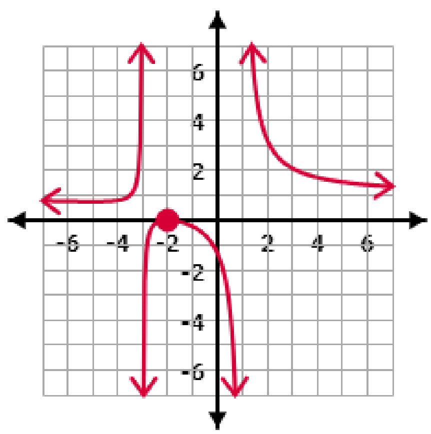
*This test review does not include practice on graphs of rates of change (position/velocity/acceleration graphs with respect to time). Be sure to review this topic from homework, quizzes, and explorations!

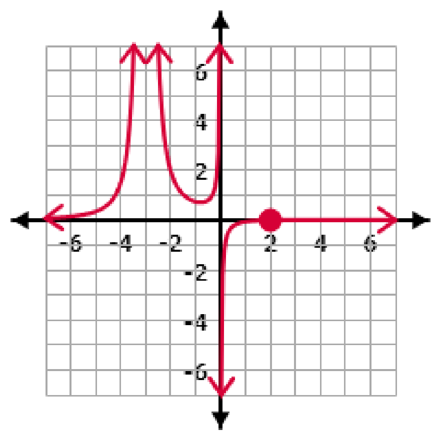
1. For each of the following functions, identify the
   - x- and y-intercepts
   - Vertical and Horizontal Asymptotes
   - Points of Discontinuity (Holes), if any

a) $\frac{6x-9}{4x^2-9}$ b) $\frac{x^2-8x-20}{2x^2+9x+10}$ c) $\frac{x^2+5x-14}{x+7}$

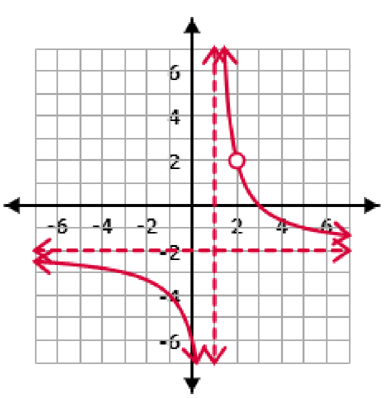
2. Write the equation of a rational function of the form $f(x)=\frac{p(x)}{q(x)}$ that creates the image of the graph below.

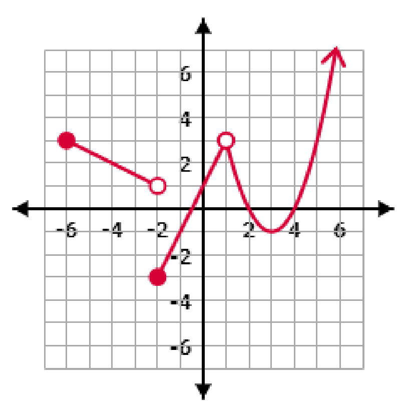






3. Evaluate the following limits given the graph of $f(x)$. Write "DNE" if the limit does not exist.





a) $\lim_{x\to 1} f(x) =$ ________ b) $\lim_{x\to 3} f(x) =$ ________

c) $\lim_{x\to 2} f(x) =$ ________ d) $\lim_{x\to \infty} f(x) =$ ________

a) $\lim_{x\to 1} f(x) =$ ________ b) $\lim_{x\to 3} f(x) =$ ________

c) $\lim_{x\to -2} f(x) =$ ________ d) $\lim_{x\to \infty} f(x) =$ ________

4. Given each function $f(x)$, use the power rule to find the derivative function, $f'(x)$.

a) $f(x) = 3x^3 - \frac{5}{2}x^2 + 4x - 1$

b) $f(x) = x + 4\sqrt{x}$

c) $f(x) = 2x^2 - \frac{1}{2x^2}$

d) $f(x) = \frac{4}{3}x\sqrt{x} + \frac{2}{x}$

5. Answer the following given $f(x) = x^2 - 7x + 3$.

a) Determine the average rate of change of $f(x)$ on the interval $[-1\,, 3]$.

b) Determine the instantaneous rate of change at $x = 2$.

c) Determine on what interval $f(x)$ is increasing and on what interval $f(x)$ is decreasing.

6. Answer the following given $f(x) = 2x^3 - 6x^2 + 3$.

a) Determine on what interval $f(x)$ is increasing and on what interval $f(x)$ is decreasing.

b) Find the coordinates of the relative extrema of $f(x)$ and identify them as local maximums or minimums.

7. Find the limit algebraically. Write "DNE" if the limit does not exist.

a) $\lim\limits_{x \to 3} \frac{x^2 - 9x - 36}{x^2 + 11x + 24}$

b) $\lim\limits_{x \to -1} \frac{x^2 - 25}{x^2 - 8x + 15}$

c) $\lim\limits_{x \to 0} \frac{\frac{1}{x+2} - \frac{1}{2}}{x}$

d) $\lim\limits_{x \to 3} \frac{\frac{1}{x^2} - \frac{1}{9}}{x - 3}$

8. A ball is tossed upward with an initial velocity of 25 m/sec from a height of 2 m. The equation for the height of the ball after t seconds is $h(t) = -5t^2 + 25t + 2$.

a) Determine the velocity function for the ball at t seconds.

b) Determine the velocity of the ball after 4 seconds.

c) Is the height of the ball increasing or decreasing at 4 seconds? Explain how you know.

d) Determine how many seconds elapse before the ball is at its maximum height. (Can you find its height, too?)

9. The table below shows the population of a small town measured every year.

Year	2012	2013	2014	2015	2016	2017
Population	500	750	1050	1500	2150	2700

a) Approximate how fast the population is growing in 2013.

b) Approximate how fast the population is growing in 2015.

OnRamps Precalculus
Unit 5 Test Review KEY

1. Identify features

a) $\frac{6x-9}{4x^2-9} = \frac{3(2x-3)}{(2x+3)(2x-3)}$

x-int: None
y-int: $(0,1)$
VA: $x = -3/2$
HA: $y = 0$
Hole: $(3/2\,,1/2)$

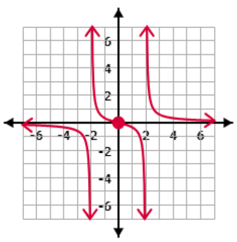
b) $\frac{x^2-8x-20}{2x^2+9x+10} = \frac{(x-10)(x+2)}{(2x+5)(x+2)}$

x-int: $(10,0)$
y-int: $(0,-2)$
VA: $x = -5/2$
HA: $y = 1/2$
Hole: $(-2\,,-12)$

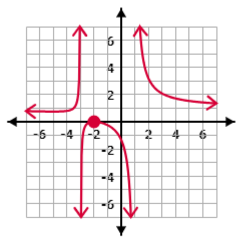
c) $\frac{x^2+5x-14}{x+7} = \frac{(x-2)(x+7)}{x+7}$

x-int: $(2,0)$
y-int: $(0,-2)$
VA: None
HA: None
Hole: $(-7\,,-9)$

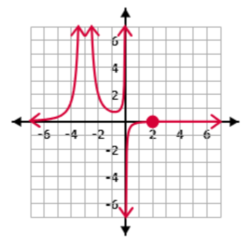
2. Write the equation.



$\frac{x}{(x+2)(x-2)}$

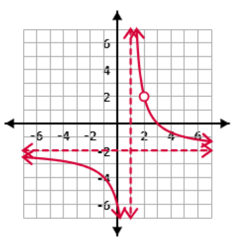


$\frac{(x+2)^2}{(x+3)(x-1)}$

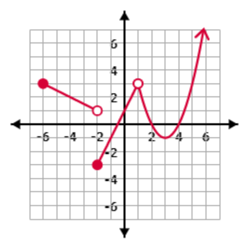


$\frac{x-2}{x(x+3)^2}$

3. Evaluate the following limits given the graph of $f(x)$. Write "DNE" if the limit does not exist.



a) $\lim_{x\to 1} f(x) =$ DNE
b) $\lim_{x\to 3} f(x) = 0$
c) $\lim_{x\to 2} f(x) = 2$
d) $\lim_{x\to \infty} f(x) = -2$



a) $\lim_{x\to 1} f(x) = 3$
b) $\lim_{x\to 3} f(x) = -1$
c) $\lim_{x\to -2} f(x) =$ DNE
d) $\lim_{x\to \infty} f(x) =$ DNE

4. Find $f'(x)$.

a) $f(x) = 3x^3 - \frac{5}{2}x^2 + 4x - 1$

$f'(x) = 9x^2 - 5x + 4$

b) $f(x) = x + 4\sqrt{x}$

$f'(x) = 1 + \frac{2}{\sqrt{x}}$

c) $f(x) = 2x^2 - \frac{1}{2x^2}$

$f'(x) = 4x + \frac{4}{x^3}$

d) $f(x) = \frac{4}{3}x\sqrt{x} + \frac{2}{x}$

$f'(x) = 2\sqrt{x} - \frac{2}{x^2}$

5. Given $f(x) = x^2 - 7x + 3$.

   a) Average rate of change on the interval $[-1\,,3]$. $\frac{f(3)-f(-1)}{3-(-1)} = \frac{-20}{4} = -5$

   b) Determine the instantaneous rate of change at $x = 2$. $f'(x) = 2x - 7, \ \therefore f'(2) = -3$

   c) Determine on what interval $f(x)$ is increasing and on what interval $f(x)$ is decreasing.

   $f'(x) = 2x - 7 = 0 \quad \Rightarrow \quad x = 7/2$

   Test x-values to the left and right of $x = 7/2 = 3.5$

   Plugging in $x = 3$, $f'(3) = -1$, so negative slope to the left of $x = 3.5$, $\therefore f(x)$ is decreasing on $(-\infty, 7/2)$.

   Plugging in $= 4$,$f'(4) = 1$, so positive slope to the right of $x = 3.5$, $\therefore f(x)$ is increasing on $(7/2, \infty)$.

6. Answer the following given $f(x) = 2x^3 - 6x^2 + 3$.

   a) Determine on what interval $f(x)$ is increasing and on what interval $f(x)$ is decreasing.

   $f'(x) = 6x^2 - 12x = 0 \quad \Rightarrow \quad 6x(x-2) = 0 \quad \Rightarrow \quad \therefore x = \{0, 2\}$

   Test x-values to the left of $x = 0$, between $x = 0$ and $x = 2$, and to the right of $x = 2$

   Plugging in $x = -1$, $f'(-1) = 18$, so positive slope to the left of $x = 0$, $\therefore f(x)$ is increasing on $(-\infty, 0)$.

   Plugging in $= 1$,$f'(1) = -6$, so negative slope between 0 and 2, $\therefore f(x)$ is decreasing on $(0, 2)$.

   Plugging in $x = 3$, $f'(3) = 18$, so positive slope to the right of $x = 2$, $\therefore f(x)$ is increasing on $(2, \infty)$.

   Hence, f is increasing on $(-\infty, 0) \cup (2, \infty)$ and decreasing on $(0, 2)$.

   b) Find the coordinates of the relative extrema of $f(x)$ and identify them as local maximums or minimums.

   $f(0) = 3$, and f changes from increasing to decreasing at $x = 0$, therefore $(0, 3)$ is a local maximum.

   $f(2) = -5$, and f changes from decreasing to increasing at $x = 2$, therefore $(2, -5)$ is a local minimum.

7. Find the limit.

   a) $\lim_{x\to3} \frac{x^2-9x-36}{x^2+11x+24} = \lim_{x\to3} \frac{(x-12)(x+3)}{(x+8)(x+3)} = \lim_{x\to3} \frac{x-12}{x+8} = \frac{3-12}{3+8} = \frac{-9}{11}$

   b) $\lim_{x\to-1} \frac{x^2-25}{x^2-8x+15} = \lim_{x-1} \frac{(x+5)(x-5)}{(x-3)(x-5)} = \lim_{x\to-1} \frac{x+5}{x-3} = \frac{-1+5}{-1-3} = \frac{4}{-4} = -1$

   c) $\lim_{x\to0} \frac{\frac{1}{x+2}-\frac{1}{2}}{x} = \lim_{x\to0} \frac{\frac{2}{2(x+2)}-\frac{x+2}{2(x+2)}}{x} = \lim_{x\to0} \frac{\frac{2-(x+2)}{2(x+2)}}{x} = \lim_{x\to0} \frac{\frac{-x}{2(x+2)}}{x} = \lim_{x\to0} \frac{-x}{2(x+2)} \cdot \frac{1}{x} = \lim_{x\to0} -\frac{1}{2x+4} = -\frac{1}{0+4} = -\frac{1}{4}$

   d) $\lim_{x\to3} \frac{\frac{1}{x^2}-\frac{1}{9}}{x-3} = \lim_{x\to3} \frac{\frac{9}{9x^2}-\frac{x^2}{9x^2}}{x-3} = \lim_{x\to3} \frac{\frac{9-x^2}{9x^2}}{x-3} = \lim_{x\to3} \frac{\frac{(3+x)(3-x)}{9x^2}}{x-3} = \lim_{x\to3} \frac{-(x+3)(x-3)}{9x^2} \cdot \frac{1}{x-3} = \lim_{x\to3} \frac{-(x+3)}{9x^2} = \frac{-(3+3)}{9(3)^2} = -\frac{2}{27}$

8. Given $h(t) = -5t^2 + 25t + 2$.

a) Determine the velocity function for the ball at t seconds.

$v(t) = h'(t) = -10t + 25$ using the power rule

b) Determine the velocity of the ball after 4 seconds.

$v(4) = -10(4) + 25 = -40 + 25 = -15,$ therefore the velocity at $t = 4$ seconds is –15 m/sec.

c) Is the height of the ball increasing or decreasing at 4 seconds? Explain how you know.

The height of the ball is decreasing at 4 seconds because the velocity is negative at that time, meaning the ball is falling down.

d) Determine how many seconds elapse before the ball is at its maximum height. (Can you find its height, too?)

Set $v(t) = 0$ because at the moment the ball changes from going up to falling down, its vertical velocity is zero (changing from positive to negative means you must cross zero).

$v(t) = -10t + 25 = 0$

$-10t = -25$

$t = \frac{-25}{-10} = \frac{5}{2} = 2.5$, so the ball reaches its maximum height after 2.5 seconds of being thrown.

To find the max height, evaluate $h(t)$ at $t = 2.5$.

$h(2.5) = -5(2.5)^2 + 25(2.5) + 2 = 33.25$, so the max height is 33.25 meters above the ground.

9. The table below shows the population of a small town measured every year.

Year	2012	2013	2014	2015	2016	2017
Population	500	750	1050	1500	2150	2700

a) Approximate how fast the population is growing in 2013.

Find the average rate of change in population from 2012 to 2014.

$\frac{\Delta\text{Population}}{\Delta\text{Year}} = \frac{1050-500}{2014-2012} = \frac{550}{2} = 225$, so the growth rate of population in 2013 is estimated to be 225 people/year.

b) Approximate how fast the population is growing in 2015.

Find the average rate of change in population from 2014 to 2016.

$\frac{\Delta\text{Population}}{\Delta\text{Year}} = \frac{2150-1050}{2016-2014} = \frac{1100}{2} = 550$, so the growth rate of population in 2015 is roughly 550 people/year.